ERAS POST-OP MONITORING · RULES ENGINE

Rule configuration: where the scoring rules are stored

Ruleset TKR 1.0.0 · 19 September 2026

> The rules live in the database, not in code. The engine in `backend/app/services/rules_engine.py` only interprets them.

The tables

TABLE	WHAT IT HOLDS
`meta.ruleset`	One row per version, e.g. TKR `1.0.0`. It holds the status (`draft` / `approved` / `retired`), the tier cut-offs (`review_at` = 5, `critical_at` = 12), `max_points_per_parameter` = 6, and who approved it. A partial unique index allows at most one approved ruleset per procedure.
`meta.procedure_parameter`	One row per ruleset × measure. It holds the question wording, when the question is asked (`collect_from_day` / `collect_to_day`), when the answer counts (`score_from_day` / `score_to_day`), the `parameter_weight`, and the rule itself in `scoring_config` (JSONB).
`meta.parameter`	The measure itself: its type (numeric / boolean / ordinal), unit and valid range, and which yes/no answer is the worrying one.
`meta.parameter_option`	Ranked options for the ordinal measures (walking aid, knee bend).

The rule logic: `scoring_config`

A band is `[from, to, points]`. It matches when `from ≤ value < to`, and a value outside every band scores 0.

Every numeric rule also states its `direction`: `higher_is_worse` (a fever rises) or `higher_is_better` (oxygen falls). It says which side of the rule is the worrying one, rather than leaving that to be guessed from where the bands sit — a guess that says nothing about a reading no band covers. The bands must agree with it: the 0-point band sits at the safe end. Yes/no and milestone measures have no direction, since “higher” means nothing for them.

`temp_c` — Scored today

```
{"direction": "higher_is_worse",
 "bands": [[0,37.5,0],[37.5,38.0,1],[38.0,38.5,2],[38.5,99,3]]}
```

Fixed bands: 0 points below 37.5 °C, rising to 3 points at 38.5 °C and above.

pain Scored today

```
{"direction": "higher_is_worse",
 "bands_by_day": {"0-3":  [[0,8,0],[8,10,1],[10,99,3]],
                  "4-10": [[0,4,0],[4,6,1],[6,8,2],[8,99,3]],
                  "11-84":[[0,3,0],[3,5,1],[5,7,2],[7,99,3]]}}
```

The bands depend on the post-op day, so pain that is expected on day 2 becomes a concern by day 12.

hr Scored today

```
{"direction": "higher_is_worse", "relative_to_baseline": true,
 "bands": [[0,10,0],[10,20,1],[20,30,2],[30,999,3]]}
```

Scored on the rise above the patient's discharge heart rate. With no discharge baseline, it scores nothing.

spo2 Scored today

```
{"direction": "higher_is_better",
 "bands": [[0,90,3],[90,93,2],[93,95,1],[95,101,0]]}
```

The one measure that worries when it falls: the 0-point band is at the top, so the direction is the other way round.

breathless Scored today

```
{"binary_score": 3, "emergency": true}
```

A "yes" is an emergency and bypasses the weighted total. A red flag at `binary_score` 3 without `emergency` (e.g. calf pain) is critical.

steps Scored today

```
{"direction": "higher_is_better",
 "trends": [{"name": "mobility_regression", "detector": "drop_from_personal_best",
             "weight": 1, "score": 2, "drop_fraction": 0.5}]}
```

A trend signal: it flags when steps fall to half of this patient's own personal best.

knee_bend Scored today

```
{"milestone": {"14": "chair", "42": "car", "84": "bike"}, "score": 2}
```

A recovery milestone: the patient is expected to reach each level by the given day. `walk_aid` uses the same shape.

Trend signals

A measure can carry `trends` alongside its bands. Each one is a separate signal with its own name and weight, so pain and `pain_rising` both appear in a nurse's reasons and are capped together against the measure they read. `direction` is what tells every detector which way is the wrong way.

```
{"direction": "higher_is_worse", "bands_by_day": { … },
 "trends": [{"name": "pain_rising", "detector": "delta_vs_rolling_mean",
             "weight": 2, "window_days": 3, "min_observations": 3,
             "bands": [[0,2,0],[2,3,1],[3,4,2],[4,99,3]]}]}
```

DETECTOR	SIGNAL IT RAISES	QUESTION IT ASKS
`delta_vs_rolling_mean`	`pain_rising`	Is today worse than this patient's own recent average, and by how much?
`reemergence`	`fever_returned`	Did this settle for a few days and then come back?
`monotonic_rise`	`slow_burn`	Has it crept steadily the wrong way without crossing a threshold?
`drop_from_personal_best`	`mobility_regression`	Has it fallen below a fraction of this patient's own best?
`streak`	`persistent_abnormal`	Has anything at all been abnormal several days running? Declared on the ruleset (`composite_config`), since it belongs to no single measure.

A detector that has not seen enough days stays silent rather than scoring 0, because 0 reads as "assessed and normal". A non-override trend must also hold across two submissions before it can raise the tier. `streak` is supported but is deliberately not switched on in the approved ruleset: at weight 1 it takes the false-positive control patients to the review cut-off, so turning it on is a calibration decision for a surgeon.

How it ties together

- **Episodes are pinned to a version.** Each patient's `txn.patient_episode.ruleset_id` fixes which version scores them, and the engine uses that rather than whatever is approved now. Approving a new version does not silently re-score patients who are already being monitored.
- **Every decision records its version.** Each `txn.check_in_evaluation` row stores the `ruleset_id` it was scored under, with its reasons, so any past alert can be explained.
- **The source of truth in the repository** is `db/seed/01_reference.sql`. Changes to a live database go through a numbered migration in `db/migrations/`. For example, `002_remove_redness_parameter.sql` removed the redness rule.
- **You can view the rules in the app** on the Rulesets screen (`/rulesets`), which is read-only by design.

Source: `db/schema/01_meta.sql`, `db/seed/01_reference.sql`, `backend/app/services/rules_engine.py`